Azure Cost Analysis — UDC-WQIS

Water Quality Information System — Infrastructure Economics

Prepared by: Oli T. Oli (Co-Founder & CTO, DAPS Analytics) **Date:** April 17, 2026 **Production URL:** https://udc-wqis.happycoast-d9b0bcde.centralus.azurecontainerapps.io/ **Azure Region:** Central US

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1. Current Azure Resource Inventory

Resource	Type	SKU / Tier	Region	Purpose
`udc-wqis-db`	Azure Database for PostgreSQL Flexible Server	Burstable B1ms (1 vCore, 2GB RAM, 32GB storage)	Central US	Primary production database
`udc-wqis-app`	Azure Container Apps	Consumption plan (scale-to-zero)	Central US	Next.js 16 application host
`udcwqisregistry`	Azure Container Registry	Basic tier	Central US	Docker image storage and versioning
`udc-wqis-jobs`	Azure Container Apps Jobs	Consumption plan	Central US	Cron jobs: USGS, EPA, WQP ingestion
`udc-wqis-storage`	Azure Blob Storage	Hot tier	Central US	Data exports, CSV uploads

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2. Monthly Cost Breakdown

Current State (April 2026)

Service	Component	Unit Cost	Monthly Cost
Azure PostgreSQL Flexible Server	Compute: Burstable B1ms, 730 hrs/mo	$0.01352/vCore/hr	$9.87
	Storage: 32GB included, 0GB overage	Included	$0.00
	Backup storage (auto 7-day)	$0.095/GB	$2.93
	PostgreSQL subtotal		**$12.80**
Azure Container Apps	vCPU: 0.5 vCPU × active hours	$0.000024/vCPU/sec	$8.20
	Memory: 1Gi × active hours	$0.000003/GiB/sec	$3.10
	Requests: ~50K/mo	$0.40/million	$0.02
	Egress: ~2GB/mo	$0.087/GB	$0.17

	HTTP scale rule overhead	Minimal	$1.11
	Container Apps subtotal		**$14.60**
Azure Container Registry	Basic tier (10GB storage included)	$5.00/mo flat	$5.00
	ACR subtotal		**$5.00**
Azure Container Apps Jobs	3 cron jobs × ~30 executions/mo	$0.000025/vCPU/sec	$0.08
	Job memory consumption	$0.000003/GiB/sec	$0.02
	Jobs subtotal		**$0.10**
Azure Blob Storage	Stored data: ~100MB	$0.018/GB	$0.002
	Operations: ~5K read/write	$0.004/10K	$0.002
	Egress: ~500MB/mo	$0.087/GB	$0.016
	Storage subtotal		**$0.02**
TOTAL			**$32.52**

Cost Summary by Time Period

Period	Cost	Notes
Daily	$1.08	$32.52 ÷ 30.25 days/mo
Weekly	$7.56	$1.08 × 7
Monthly	$32.52	Base production cost
Annual	$390.24	$32.52 × 12
3-Year	$1,170.72	Without scaling
5-Year	$1,951.20	Without scaling

> ***Context:*** *At $32.52/month, UDC-WQIS costs less than most individual SaaS subscriptions while running a production-grade environmental monitoring platform ingesting 200,000+ real government data points.*

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3. Database Growth Projections

Current State

Metric	Value
Current database size	~46 MB
Total rows	216,000+
Primary tables	`readings`, `measurements`, `stations`, `parameters`, `ingestion_log`

Storage tier	32 GB included in B1ms
Storage used (%)	0.14%

Daily Ingestion Volume

Source	Frequency	Estimated Records/Day	Notes
USGS NWIS	Daily at 06:00 UTC	9,120	7 sites × 6 parameters × 8 readings (last 24h at 3h intervals)
EPA Water Quality Portal	Daily at 07:00 UTC	~200	Lab results; varies with sampling activity
WQP Extended Parameters	Weekly Monday 08:00 UTC	~~500/week (~~71/day avg)	Nutrients, metals, emerging contaminants
Manual uploads	On demand	Variable	Faculty and researcher CSV imports
Total estimated		**~9,400/day**	~282,000/month at current activity level

Storage Growth Model

Timeframe	Estimated Row Count	Estimated DB Size	% of 32GB Limit
Now (April 2026)	216,000	46 MB	0.14%
6 months (Oct 2026)	1,900,000	400 MB	1.25%
1 year (Apr 2027)	3,600,000	760 MB	2.38%
2 years (Apr 2028)	7,000,000	1.5 GB	4.69%
3 years (Apr 2029)	10,400,000	2.2 GB	6.88%
6 years (Apr 2032)	20,700,000	4.4 GB	13.75%
~28 years	~150,000,000	32 GB	100% — upgrade needed

Key finding: The current 32GB storage allocation is sufficient for **approximately 28 years of ingestion at current rates** for a single watershed deployment. Storage is not a near-term concern. CPU and memory during peak ingestion windows are the more relevant scaling considerations.

Storage Overage Costs (if triggered)

Scenario	Additional Storage	Additional Cost
Upgrade to 64GB	+32GB	+$3.84/mo
Upgrade to 128GB	+96GB	+$11.52/mo
Upgrade to 512GB	+480GB	+$57.60/mo

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4. Ingestion Volume Analysis

USGS NWIS — Primary Data Source

USGS Station	Site ID	Status	Parameters	Readings/Day
NW Branch Anacostia at Riverdale	01651000	Active	5 (temp, DO, pH, turbidity, conductivity)	40
NE Branch near Riverdale	01649500	Active	5	40
Hickey Run at National Arboretum	01651827	Active	5	40
NW Branch above Anacostia R	01651750	Active	5	40
Anacostia R at US Hwy 50	01651800	Active	5	40
Watts Branch at Washington	01651770	Active	5	40
Unnamed tributary near Hyattsville	01651730	Active	5	40
Total USGS	7 sites		35 streams/day	**280 readings/day**

> *Note: USGS 15-minute interval data covers the prior 24 hours, yielding 96 timestamps × 5 parameters × 7 sites = up to 3,360 potential raw readings per ingestion run. After deduplication (UNIQUE index + ON CONFLICT upsert), net new rows per day approximate 280–500 depending on station uptime.*

EPA and WQP

Source	Trigger	Approx Records	Type
EPA WQP (DC FIPS US:11)	Daily	50–500	Lab-analyzed: nutrients, metals, biological
WQP Extended (HUC 02070010)	Weekly	200–1,000	Emerging contaminants, extended parameter set

EPA and WQP ingestion volumes are highly variable — they depend on when agencies upload lab results to the federal portal. Ingestion runs that return zero records are normal and expected.

Total Ingestion Cost (Azure Functions / Container Apps Jobs)

Azure charges for Container Apps Jobs execution at approximately $0.000025/vCPU-second. Each ingestion job runs for roughly 30–60 seconds at 0.5 vCPU:

Job	Frequency	Duration	vCPU-seconds/mo	Monthly Cost
USGS ingest	Daily	45 sec	675	$0.017

EPA ingest	Daily	30 sec	450	$0.011
WQP ingest	Weekly	60 sec	120	$0.003
Total			**1,245**	**$0.031**

Ingestion infrastructure is effectively free at current scale.

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5. Cost Optimization Options

Option A: Scale-to-Zero Container App (Already Enabled)

The Container App is currently configured with `min-replicas: 0`, meaning it scales to zero when no requests arrive. This saves approximately $8–12/month compared to maintaining a permanently running minimum replica.

Configuration	Monthly Cost	Savings
min-replicas: 0 (current)	~$14.60	Baseline
min-replicas: 1 (always-on)	~$24.80	-$10.20/mo
min-replicas: 1 (business hours only, ~10 hrs/day)	~$18.40	-$3.80/mo

Recommendation: Keep scale-to-zero for current single-tenant deployment. Add `min-replicas: 1` only when signed paying clients require uptime SLA guarantees.

Option B: Downgrade Container Registry

Tier	Cost	Storage	Included Features
Basic (current)	$5.00/mo	10 GB	Pull from Container Apps
Standard	$20.00/mo	100 GB	Geo-replication, enhanced throughput
Premium	$50.00/mo	500 GB	Private endpoints, content trust

Recommendation: Basic tier is appropriate for current single-image, single-region deployment. No change needed.

Option C: Managed Identity / Key Vault (Security Upgrade, Minimal Cost)

Adding Azure Key Vault for secrets management costs approximately $0.03/secret/month. For 5 secrets (DATABASE_URL, ANTHROPIC_API_KEY, ADMIN_API_KEY, INGEST_API_KEY, APP_URL):

Addition	Monthly Cost
Azure Key Vault (5 secrets)	$0.15
Managed Identity (free)	$0.00
Total addition	**$0.15/mo**

This is the highest-priority security upgrade before any government partnership discussions.

Option D: Azure for Startups Credits

Applying for Microsoft for Startups Founders Hub credits can eliminate Azure costs entirely for 12 months:

Credit Level	Coverage at $32.52/mo	Duration
$5,000	153 months (12+ years at current scale)	N/A — use for growth
$10,000	Full Year 1 scaling to 10 watersheds	12 months
$25,000	Full Year 1 + Azure OpenAI + Azure AI Search	12 months

Recommendation: Apply immediately. The application narrative is documented in `AZURE_STARTUP_APPLICATION.md`.

Option E: Reserved Instances (Future)

Once the deployment is stable and client-committed for 12+ months, reserved instance pricing for PostgreSQL provides significant savings:

Commitment	PostgreSQL Compute Cost	Savings vs. Pay-as-you-go
No reservation (current)	$9.87/mo	Baseline
1-year reserved	$6.37/mo	-35%
3-year reserved	$4.50/mo	-54%

Recommendation: Pursue once first paying client is signed.

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6. Platform Comparison: Vercel vs. Azure

Factor	Vercel Hobby (Previous)	Azure (Current)
Monthly cost	$0	$32.52
Annual cost	$0	$390.24
Cron jobs	1 maximum (Hobby plan hard limit)	Unlimited (Container Apps Jobs)
Function timeout	60 seconds	10 minutes (Container Apps Jobs)
Database	Neon PostgreSQL serverless	Azure PostgreSQL Flexible Server
SLA	None (Hobby)	99.9% (Container Apps + PostgreSQL)
Custom SSL	Vercel-managed only	Azure-managed + BYO certificate
FedRAMP authorization	No	Yes (Azure Government pathway)
Secrets management	Environment variables	Azure Key Vault (available)
Monitoring	Vercel dashboard only	Azure Monitor + App Insights
Multi-region	Vercel Edge Network	Azure multi-region (available)
Government procurement	Not viable	Direct pathway via Azure Government

Startup credits	Vercel credits (limited)	Microsoft Founders Hub ($5K–$150K)
White-label isolation	Project-per-deployment	Container App per client (clean)
Log retention	1 day (Hobby)	90 days (Log Analytics)

Summary: Vercel Hobby was appropriate for development and proof-of-concept. Azure is required for production-grade deployment, government partnerships, multi-tenant white-label scaling, and grant compliance. The $32.52/month cost differential is justified at the first paying client contract.

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7. White-Label Scaling Cost Projections

Per-Watershed Marginal Cost

Each additional white-label watershed deployment adds the following Azure resources:

Resource	Per-Watershed Addition	Monthly Cost
Container App (new instance)	Consumption plan	~$10–15
PostgreSQL database	Shared server, new database	~$2 (storage)
Container Apps Jobs (3 cron jobs)	Timer triggers	~$0.03
Custom domain SSL	Azure-managed	$0
Per-watershed marginal cost		**~$12–17/mo**

> *Note: PostgreSQL Flexible Server supports multiple databases within a single server. Additional watersheds share compute costs until the server is CPU-constrained (~10+ active concurrent clients), at which point a second server is warranted.*

Revenue vs. Cost at Scale

Deployments	Monthly Revenue ($500/watershed)	Monthly Azure Cost	Gross Margin	Margin %
1 (current, UDC)	$0 (partnership)	$32.52	-$32.52	—
5 paying clients	$2,500	$32.52 + (5 × $15) = $107.52	$2,392.48	95.7%
10 paying clients	$5,000	$32.52 + (10 × $15) = $182.52	$4,817.48	96.4%
25 paying clients	$12,500	$32.52 + (25 × $15) = $407.52	$12,092.48	96.7%
50 paying clients	$25,000	$32.52 + (50 × $15) = $782.52	$24,217.48	96.9%
100 paying clients	$50,000	$32.52 + (100 × $15) = $1,532.52	$48,467.48	96.9%

Key finding: Infrastructure gross margins stabilize above 96% at scale due to Container Apps Consumption pricing — costs scale linearly with usage, not in step-function jumps. There is no over-provisioning tax.

Infrastructure Cost at 10, 50, and 100 Watersheds

10 Watersheds

Service	Cost	Notes
PostgreSQL Flexible Server (B2s, 2 vCore)	$26.00	Upgrade from B1ms at ~7 clients
Container Apps (10 apps)	$120.00	~$12/app/mo average
Container Registry (Standard)	$20.00	Higher throughput for 10 deployments
Container Apps Jobs (30 cron jobs)	$0.30	3 per watershed
Storage + monitoring	$5.00	Blob storage, basic monitoring
Total	**$171.30/mo**	$17.13/watershed/mo

50 Watersheds

Service	Cost	Notes
PostgreSQL Flexible Server (GP D4s, 4 vCore)	$150.00	General Purpose tier at this scale
Container Apps (50 apps)	$600.00	~$12/app/mo
Container Registry (Standard)	$20.00	Unchanged
Container Apps Jobs (150 cron jobs)	$1.50	3 per watershed
Azure Monitor + App Insights	$25.00	Per-deployment alerting
Storage + egress	$20.00	
Total	**$816.50/mo**	$16.33/watershed/mo

100 Watersheds

Service	Cost	Notes
PostgreSQL (2× GP D4s servers)	$300.00	Sharded for isolation
Container Apps (100 apps)	$1,200.00	
Container Registry (Premium)	$50.00	Private endpoints, content trust
Container Apps Jobs (300 cron jobs)	$3.00	
Azure Monitor	$50.00	
Azure Key Vault (per-tenant secrets)	$15.00	
Storage + egress	$40.00	
Total	**$1,658.00/mo**	$16.58/watershed/mo

Conclusion: Infrastructure cost per watershed **decreases with scale** and stabilizes at approximately $16–17/watershed/month — a 96.6% gross margin on the $500/month subscription price. This is a highly capital-efficient SaaS model.

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8. Cost Sensitivity Analysis

Variable	Change	Monthly Cost Impact
Add Azure Key Vault	+5 secrets	+$0.15
Add Azure Monitor + App Insights	Basic tier	+$0 (free tier)
Enable min-replicas: 1 (always-on)	Container App uptime SLA	+$10.20
Add Azure OpenAI (GPT-4o, 1M tokens/mo)	AI assistant migration	+$15–30
Add Azure AI Search (Basic, 1 index)	Research RAG	+$25
Upgrade PostgreSQL to General Purpose	2 vCore, 8GB (needed at ~7 clients)	+$66
Enable geo-redundant backups	PostgreSQL option	+$6.50
Add Azure CDN	Global edge caching	+$2–5

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9. Recommended Immediate Actions

Priority	Action	Monthly Impact	Effort
1	Apply for Microsoft Founders Hub credits	-$32.52 (free)	2 hours
2	Add Azure Key Vault for secrets	+$0.15	1 hour
3	Enable Azure Monitor health alert on `/api/health`	+$0	30 min
4	Set up Log Analytics workspace (90-day retention)	+$0	30 min
5	Enable 1-year Reserved Instance on PostgreSQL (post-client)	-$3.50	15 min

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